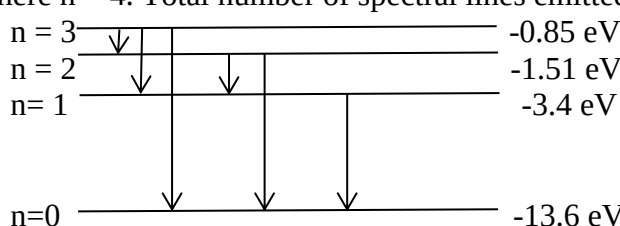
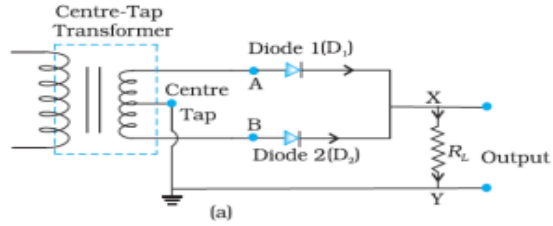
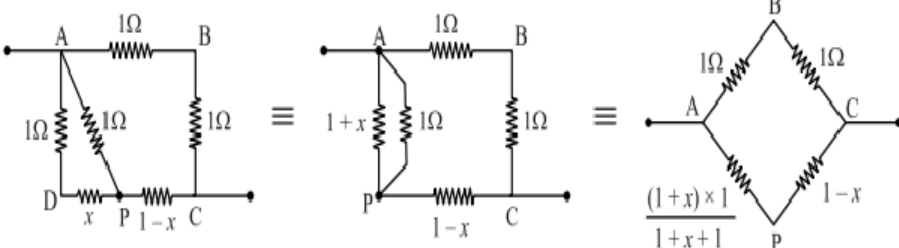
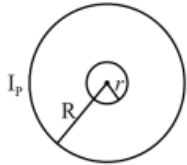
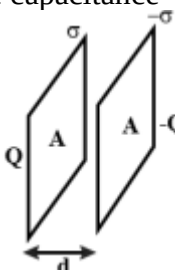
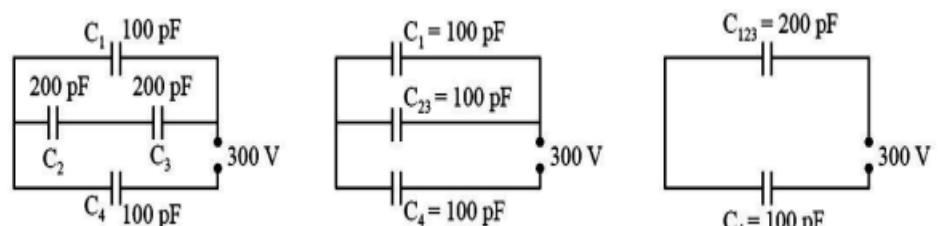
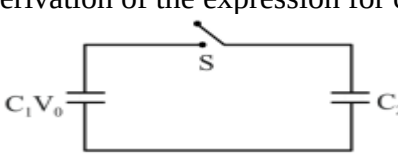


Q. NO.	MARKING SCHEME (PRE-BOARD) CLASS – XII PHYSICS SECTION A	MARKS
1.	B.	1
2.	C.	1
3.	A.	1
4.	B.	1
5.	C.	1
6.	D.	1
7.	B.	1
8.	C.	1
9.	A.	1
10.	C.	1
11.	D.	1
12.	B.	1
13.	D.	1
14.	A.	1
15.	A.	1
16. .	B	1
SECTION B		
17.	a) when $\theta = 0^\circ$ or 180° $F = qvB \sin\theta = 0$ When electron moves parallel or antiparallel to the direction of magnetic field, it suffer no deflection. b) $r = mv/qB$ as $B_2 > B_1$ $r_Q < r_p$ the proton Q describe circular path of smaller radius	1/2 1/2 1/2 1/2
18.	(i) X is diamagnetic and Y is paramagnetic material (ii) when a diamagnetic bar is placed in magnetic field, the field lines are repelled or expelled and the field inside the material is reduced. When a paramagnetic bar is placed in a magnetic field, the field lines get concentrated inside the material and hence the field is increased. Or As the susceptibility is positive and small, the magnetic material is paramagnetic material. Any two properties.	1/2+1/2 1/2+1/2 1 1
19.	(a) Gamma rays, any one use. (b) Infrared rays, any one use.	1 1
20.	(a) here $n = 4$. Total number of spectral lines emitted = $n(n-1)/2 = 4 \times 3/2 = 6$  (b) the spectral line of shortest wavelength will be emitted in the transition $n = 4$ to $n = 1$ as it correspond to the photon of maximum energy.	1/2 1 1/2

21.	(a) Rectifier (b) Circuit diagram of full wave rectifier  (a)	1 1
SECTION C		
22.	 As point B and P are at same potential, the bridge is balanced and hence $\frac{1}{1} = \frac{(1+x)}{(2+x)} \Rightarrow x = (\sqrt{2} - 1) \text{ ohm}$	1.5 1.5
23.	(a) Consider the case $r > a$. The Amperian loop, labelled 2, is a circle concentric with the cross-section. For this loop, $L = 2 \pi r$ Using Ampere circuital Law, we can write, $B (2\pi r) = \mu_0 I$ $B = \mu_0 I / 2\pi r$ $B \propto 1/r \quad (r > a)$ (b) Consider the case $r < a$. The Amperian loop is a circle labelled 1. For this loop, taking the radius of the circle to be r, $L = 2 \pi r$. Now the current enclosed I_e is not I, but is less than this value. Since the current distribution is uniform, the current enclosed is, $I_e = I (\pi r^2 / \pi a^2) = I (r^2 / a^2)$ Using Ampere circuital Law, we can write, $B (2\pi r) = \mu_0 I (r^2 / a^2)$ $B = (\mu_0 I / 2\pi a^2) r$ $B \propto r \quad (r < a)$ Or Principle of moving coil galvanometer: A current carrying coil placed in a magnetic field experiences a current dependent torque, which tends to rotate the coil and produce angular deflection. (a) Radial magnetic field maximize the torque on the coil by making $\tau = NIAB \sin 90^\circ = NIAB$ and provide linear current scale. (b) A cylindrical soft iron core makes the field radial. This increase the strength of magnetic field and hence increase the sensitivity of galvanometer. A galvanometer as such cannot be used to measure current in the circuit due to two reasons. (i) Galvanometer can only detect small currents and its coil is likely to be damaged by currents of (mA/A) range. (ii) For measuring current it has to be first calibrated.	1.5 1/2 1 1/2 1
24.	Mutual Induction: Mutual induction is the phenomenon of production of induced emf in one coil due to a change of current in the neighbouring coil. SI unit of mutual induction is Henry.	1/2 1/2

	 Let a current I_p flow through the circular loop of radius R. The magnetic induction at the centre of the loop is $B_p = \mu_0 I_p / 2R$ As, $r \ll R$, the magnetic induction B_p may be considered to be constant over the entire cross sectional area of inner loop of radius r. Hence magnetic flux linked with the smaller loop will be $\phi_s = B_p A_s = (\mu_0 I_p / 2R) \pi r^2$ $\phi_s = M I_p$ $M = (\mu_0 / 2R) \pi r^2$	1/2 1/2 1/2 1/2
25.	For a LCR series circuit $\langle P \rangle = \langle v_i \rangle = \langle (v_m \sin \omega t)(i_m \sin \omega t + \phi) \rangle = (v_m i_m \cos \phi) / 2$ For resonance $X_L = X_C$ $\omega L = 1 / \omega C$ $\omega = 1 / \sqrt{LC}$	1.5 1/2 1
26.	Correct diagram $M = -f_o / f_e$ Any two advantages	1.5 1/2 1/2+1/2
27.	Here $\lambda = 2\text{nm}$ a) both electron and photon have same momentum. i.e. $p = h / \lambda = 3.315 \times 10^{-25} \text{ Kgms}^{-1}$ b) energy of photon $E = hc / \lambda = 9.945 \times 10^{-17} \text{ J} = (9.945 \times 10^{-17} / 1.6 \times 10^{-19}) = 6.22 \times 10^2 \text{ eV}$ c) Kinetic energy of electron $= p^2 / 2m = 0.377 \text{ eV}$	1 1 1
28.	(i) Threshold wavelength $\lambda_0 = c / \nu_0$ As $\nu_0(Q) > \nu_0(P)$ $\lambda_0(Q) < \lambda_0(P)$ thus metal Q has smaller threshold wavelength. (ii) According to Einstein's photoelectric equation $hc / \lambda = hc / \lambda_0 + \text{kinetic energy of photoelectron}$ for the same λ of incident radiation, LHS is constant. So metal Q with smaller value of λ_0 will emit photoelectron of smaller kinetic energy. (iii) Stopping potential V_0 will remain same because it is independent of intensity and hence of distance between the light source and metal surface.	1 1 1
SECTION D		
29.	(i) d (ii) b (iii) c or b (iv) d	1 1 1 1
30.	(i) a (ii) a (iii) b (iv) b or b	1 1 1 1
SECTION E		
31.	Drift velocity: It is the average velocity acquired by the free electrons superimposed over the random motion in the direction opposite to electric field and along the length of the	1/2

	metallic conductor. Derivation $I = ne A V_d$ Here, $I = I_1 + I_2 \dots$ (i) Let V = Potential difference between A and B. For cell E_1 Then, $V = \mathcal{E}_1 - I_1 r_1$ $I_1 = (E_1 - V)/r_1$ Similarly for cell $\mathcal{E}_2$ Then, $V = \mathcal{E}_2 - I_2 r_2$ $I_2 = (E_2 - V)/r_2$ $I = (E_1 - V)/r_1 + (E_2 - V)/r_2$ $I = (E_1/r_1 + E_2/r_2) - V(1/r_1 + 1/r_2)$ $V = \{(E_1 r_2 + E_2 r_1)/(r_1 + r_2)\} - I(r_1 r_2 / (r_1 + r_2))$ Comparing the above equation with the equivalent circuit of emf 'E_{eq}' and internal resistance 'r_{eq}' then, $V = E_{eq} - I r_{eq}$ (i) $E_{eq} = \{(E_1 r_2 + E_2 r_1)/(r_1 + r_2)\}$ (ii) $r_{eq} = (r_1 r_2 / (r_1 + r_2))$ (iii) The potential difference between A and B, $V = E_{eq} - I r_{eq}$ Or (a) $i_2 = i_1 + i_3 + i_4 + i_5$ Definition (b) Next, take a closed loop, say ABCCEA, and apply Kirchhoff's second law: $-IR - (1/2)IR + \text{emf} = 0$. Where R is the resistance of each edge and the emf of battery. Thus, $\text{emf} = (5/2) IR$ The equivalent resistance R of the network is $= \text{emf}/3I = (5/6)R$. For $R = 1 \text{ ohm}$, $R = (5/6) \text{ ohm}$ and for $\text{emf} = 10 \text{ V}$, the total current ($= 3I$) in the network is $3I = 10V/(5/6) = 12 \text{ A}$. Hence, the total current I in the circuit external to the network is : 12 A.	1.5 3 2 3
32.	(i) A wavefront is defined as a surface of constant phase. (a) The ray indicates the direction of propagation of wave while the wavefront is the surface of constant phase. (b) The ray at each point of a wavefront is normal to the wavefront at that point. (ii) AB: Incident Plane Wave Front & CE is Refracted Wave front . $\sin i = BC/AC$ & $\sin r = AE/AC$ $\sin i / \sin r = BC/AE = v_1/v_2 = \text{constant}$ (iii) $\Theta = \lambda/a$ i.e. $a = \lambda/\Theta = 6 \times 10^{-7}/0.1 \times \pi/180 = 304 \times 10^{-4} \text{ m}$	1 2

	(iv) Two differences between interference pattern and diffraction pattern Or (a) Correct derivation (b) correct answers 7.5cm and 48cm respectively	1 1 3 2
33.	(i) Derivation of the expression for the capacitance  Let the two plates be kept parallel to each other separated by a distance d and cross-sectional area of each plate is A. Electric field by a single thin plate $E = \sigma / 2\epsilon_0$ Total electric field between the plates $E = \sigma / \epsilon_0 = Q/A \epsilon_0$ Potential difference between the plates $V = Ed = [Q/A \epsilon_0] d$. Capacitance $C = Q/V = A\epsilon_0 / d$ (ii)  The equivalent capacitance = $200/3$ pF charge on $C_4 = 200/3 \times 10^{-12} \times 300 = 2 \times 10^{-8}$ C potential difference across $C_4 = 200 \times 10^{-12} \times 300 / 3 \times 100 \times 10^{-12} = 200$ V potential difference across $C_1 = 300 - 200 = 100$ V charge on $C_1 = 100 \times 10^{-12} \times 100 = 1 \times 10^{-8}$ C potential difference across C_2 and C_3 series combination = 100 V potential difference across C_2 and C_3 each = 50 V charge on C_2 and C_3 each = $200 \times 10^{-12} \times 50 = 1 \times 10^{-8}$ C OR (i) Derivation of the expression for capacitance with dielectric slab ($t < d$)  (ii) Before the connection of switch S, Initial energy $U_i = 1/2 C_1 V_0^2 + 1/2 C_2 0^2 = 1/2 C_1 V_0^2$ After the connection of switch S common potential $V = (C_1 V_1 + C_2 V_2) / (C_1 + C_2) = C_1 V_0 / (C_1 + C_2)$ Final energy = $U_f = 1/2 (C_1 + C_2) (C_1 V_0)^2 / (C_1 + C_2)^2$ $U_f : U_i = C_1 / (C_1 + C_2)$	2 1 1/2 1/2 1/2 1/2 1/2 3 1/2 1/2 1/2 1/2